

Problem Set 1 – Solutions

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Q1

Household maximization problem

$$\max_{\{c_t, k_{t+1}, b_{t+1}, l_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\sigma}}{1-\sigma} - \chi \frac{l_t^{1+\phi}}{1+\phi} \right]$$

$$\text{s.t. } c_t + k_{t+1} + b_{t+1} = (1 + r_t^k - \delta)k_t + (1 + r_t^b)b_t + w_t l_t - T_t$$

Lagrangian

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\sigma}}{1-\sigma} - \chi \frac{l_t^{1+\phi}}{1+\phi} + \lambda_t ((1 + r_t^k - \delta)k_t + (1 + r_t^b)b_t + w_t l_t - T_t - c_t - k_{t+1} - b_{t+1}) \right]$$

Q2. FOCs

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0 \quad : \quad c_t^{-\sigma} = \lambda_t$$

$$\frac{\partial \mathcal{L}}{\partial l_t} = 0 \quad : \quad \chi l_t^\phi = \lambda_t w_t \Rightarrow \boxed{\chi l_t^\phi = c_t^{-\sigma} w_t}$$

$$\frac{\partial \mathcal{L}}{\partial k_{t+1}} = 0 \quad : \quad \lambda_t = \beta \mathbb{E}_t [\lambda_{t+1} (1 + r_{t+1}^k - \delta)] \Rightarrow \boxed{c_t^{-\sigma} = \beta \mathbb{E}_t [c_{t+1}^{-\sigma} (1 + r_{t+1}^k - \delta)]}$$

$$\frac{\partial \mathcal{L}}{\partial b_{t+1}} = 0 \quad : \quad \lambda_t = \beta \mathbb{E}_t [\lambda_{t+1} (1 + r_{t+1}^b)] \Rightarrow \boxed{c_t^{-\sigma} = \beta \mathbb{E}_t [c_{t+1}^{-\sigma} (1 + r_{t+1}^b)]}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \quad : \quad \boxed{c_t + k_{t+1} + b_{t+1} = (1 + r_t^k - \delta)k_t + (1 + r_t^b)b_t + w_t l_t - T_t}$$

Transversality conditions

$$\boxed{\lim_{t \rightarrow \infty} \mathbb{E}_0 [\beta^t u_c(c_t, l_t) k_{t+1}] = 0}$$

$$\boxed{\lim_{t \rightarrow \infty} \mathbb{E}_0 [\beta^t u_c(c_t, l_t) b_{t+1}] = 0}$$

Q3. Goods market clearing

According to Walras' law, if $n - 1$ markets clear, then the n -th market also clears. Therefore, we will use the other market-clearing conditions (labour, capital, and bonds) together to derive the final goods market-clearing condition.

Household BC: $c_t + k_{t+1} + b_{t+1} = (1 + r_t^k - \delta)k_t + (1 + r_t^b)b_t + w_t l_t - T_t$

Firms zero profit condition: under perfect competition, firms pay factors their marginal products, so total output equals factor payments: $y_t = r_t^k k_t + w_t l_t$

Substituting into the household BC:

$$\begin{aligned} c_t + k_{t+1} + b_{t+1} &= (1 - \delta)k_t + (1 + r_t^b)b_t + y_t - T_t \\ c_t + k_{t+1} - (1 - \delta)k_t + b_{t+1} - (1 + r_t^b)b_t &= y_t - T_t \end{aligned}$$

The government BC is:

$$\begin{aligned} b_{t+1} &= (1 + r_t^b)b_t + g_t - T_t \\ g_t &= b_{t+1} - (1 + r_t^b)b_t + T_t \end{aligned}$$

Substituting the government BC into the previous equation we obtain the final good market clearing:

$$y_t = c_t + k_{t+1} - (1 - \delta)k_t + g_t$$

This is the resource constraint of the economy. Total output is allocated to: private consumption (c_t), investment ($k_{t+1} - (1 - \delta)k_t$) and government spending (g_t).

Q4. Competitive Equilibrium

Conditional on the exogenous processes $\{\varepsilon_{g,t}, \varepsilon_{z,t}\}$, a competitive equilibrium is a set of sequences $\{c_t, k_{t+1}, b_{t+1}, y_t, w_t, r_t^k, r_t^b, l_t, g_t, z_t, T_t\}_{t=0}^{\infty}$ such that, for all t :

(1) Intratemporal FOC (Labor supply): $\chi l_t^\phi = c_t^{-\sigma} w_t$

(2) Euler for capital: $c_t^{-\sigma} = \beta \mathbb{E}_t [c_{t+1}^{-\sigma} (1 + r_{t+1}^k - \delta)]$

(3) Euler for bonds: $c_t^{-\sigma} = \beta \mathbb{E}_t [c_{t+1}^{-\sigma} (1 + r_{t+1}^b)]$

(4) Production: $y_t = z_t k_t^\alpha l_t^{1-\alpha}$

Firms maximize profits taking prices as given:

$$\Pi_t = \max_{k_t, l_t} z_t k_t^\alpha l_t^{1-\alpha} - r_t^k k_t - w_t l_t$$

$$\frac{\partial \Pi_t}{\partial l_t} = 0 \Leftrightarrow (1 - \alpha) z_t k_t^\alpha l_t^{-\alpha} = w_t \quad \frac{\partial \Pi_t}{\partial k_t} = 0 \Leftrightarrow \alpha z_t k_t^{\alpha-1} l_t^{1-\alpha} = r_t^k$$

(5) Wage: $w_t = (1 - \alpha)z_t k_t^\alpha l_t^{1-\alpha} = (1 - \alpha) \frac{y_t}{l_t}$

(6) Rental rate: $r_t^k = \alpha z_t k_t^{\alpha-1} l_t^{1-\alpha} = \alpha \frac{y_t}{k_t}$

(7) Government BC: $b_{t+1} = (1 + r_t^b)b_t + g_t - T_t$

(8) Fiscal rule: $T_t = \bar{T} \left(\frac{b_t}{\bar{b}} \right)^{\phi_{T,b}}$

(9) Resource constraint: $y_t = c_t + k_{t+1} - (1 - \delta)k_t + g_t$

The exogenous states follow:

(10) $\log z_t = (1 - \rho_z) \log \bar{z} + \rho_z \log z_{t-1} + \varepsilon_{z,t}$

(11) $\log g_t = (1 - \rho_g) \log \bar{g} + \rho_g \log g_{t-1} + \varepsilon_{g,t}$

At time t , there are two types of variables:

- Exogenous variables: $\varepsilon_{g,t}$ and $\varepsilon_{z,t}$
- Endogenous variables: $c_t, k_{t+1}, b_{t+1}, y_t, w_t, r_t^k, r_t^b, l_t, g_t, z_t$ and T_t . Note that b_t and k_t are given for each period t .

There are 11 equations and 11 endogenous variables; thus, the model is identified.

Q5. Computation of steady-state values

In steady state, all variables are constant and $\bar{z} = 1$. The Euler equations imply:

$$\begin{aligned} 1 &= \beta(1 + \bar{r}^b) \\ 1 &= \beta(1 + \bar{r}^k - \delta) \end{aligned}$$

With $\beta = 1$ and $\delta = 0.025$:

$$\bar{r}^b = 0 \quad \text{and} \quad \bar{r}^k = \delta = 0.025$$

From the firm's FOC for capital, $\bar{r}^k = \alpha \bar{y} / \bar{k}$, so:

$$\frac{\bar{k}}{\bar{y}} = \frac{\alpha}{\bar{r}^k}$$

The target labour supply is $\bar{\ell} = 1/3$, so from the rental rate we obtain:

$$\bar{k} = \bar{\ell} \left(\frac{\alpha}{\bar{r}^k} \right)^{\frac{1}{1-\alpha}} = \frac{1}{3} \left(\frac{0.36}{0.025} \right)^{\frac{1}{0.64}} \implies \bar{k} \approx 21.5189$$

Then:

$$\bar{y} = \bar{k}^\alpha \bar{\ell}^{1-\alpha} \implies \boxed{\bar{y} \approx 1.49437}$$

$$\bar{w} = (1 - \alpha) \frac{\bar{y}}{\bar{\ell}} \implies \boxed{\bar{w} \approx 2.86919}$$

Government spending is 10% of output, so $\bar{g} = 0.1 \bar{y} \approx 0.149437$.

Government debt is 60% of annual output (240% of quarterly output), so $\boxed{\bar{b} = 2.4 \bar{y} \approx 3.58648}$.

The steady-state government budget constraint is $\bar{b} = (1 + \bar{r}^b) \bar{b} + \bar{g} - \bar{T}$, which, because of $\bar{r}^b = 0$, implies:

$$\boxed{\bar{T} = \bar{g} \approx 0.149437}$$

From the resource constraint, $\bar{y} = \bar{c} + \delta \bar{k} + \bar{g}$, so:

$$\bar{c} = \bar{y} - \delta \bar{k} - \bar{g} \approx 1.49437 - 0.53797 - 0.14937 \implies \boxed{\bar{c} \approx 0.80696}$$

Finally, using the steady-state intratemporal condition to pin down χ : $\chi \bar{\ell}^\phi = \bar{w} \bar{c}^{-\sigma}$:

$$\chi = \frac{\bar{w}}{\bar{c}^\sigma \bar{\ell}^{0.5}} \implies \boxed{\chi \approx 6.15840}$$

Q6. Log-linearization

For any variable with positive steady state, we define $\hat{x}_t \equiv \log\left(\frac{x_t}{\bar{x}}\right) \approx \frac{x_t - \bar{x}}{\bar{x}}$.

The log-linearised system is:

(1) Intratemporal FOC (Labor supply).

Taking logs of $\chi \ell_t^\phi = c_t^{-\sigma} w_t$ at time t and at the steady state:

$$\log \chi + \phi \log \ell_t = -\sigma \log c_t + \log w_t \tag{I}$$

$$\log \chi + \phi \log \bar{\ell} = -\sigma \log \bar{c} + \log \bar{w} \tag{II}$$

Subtracting (II) from (I):

$$\boxed{\phi \hat{\ell}_t + \sigma \hat{c}_t = \hat{w}_t}$$

(2) Euler equation for capital.

Taking logs of $c_t^{-\sigma} = \beta E_t [c_{t+1}^{-\sigma} (1 + r_{t+1}^k - \delta)]$:

$$-\sigma \log c_t = \log \beta - \sigma E_t \log c_{t+1} + E_t \log (1 + r_{t+1}^k - \delta) \tag{III}$$

$$-\sigma \log \bar{c} = \log \beta - \sigma \log \bar{c} + \log (1 + \bar{r}^k - \delta) \implies 1 + \bar{r}^k - \delta = \beta^{-1} = 1 \tag{IV}$$

Subtracting (IV) from (III):

$$-\sigma \hat{c}_t = E_t \left[-\sigma \hat{c}_{t+1} + \log \frac{1 + r_{t+1}^k - \delta}{1 + \bar{r}^k - \delta} \right]$$

Since:

$$\frac{1 + r_{t+1}^k - \delta}{1 + \bar{r}^k - \delta} = 1 + \frac{r_{t+1}^k - \bar{r}^k}{1 + \bar{r}^k - \delta} = 1 + \bar{r}^k \hat{r}_{t+1}^k$$

Then:

$$-\sigma \hat{c}_t = E_t[-\sigma \hat{c}_{t+1} + \log(1 + \bar{r}^k \hat{r}_{t+1}^k)]$$

$$\boxed{-\sigma \hat{c}_t = E_t[-\sigma \hat{c}_{t+1} + \bar{r}^k \hat{r}_{t+1}^k]}$$

(3) Euler equation for bonds.

Taking logs of $c_t^{-\sigma} = \beta E_t[c_{t+1}^{-\sigma}(1 + r_{t+1}^b)]$:

$$-\log \beta = \sigma(\log c_t - E_t \log c_{t+1}) + E_t \log(1 + r_{t+1}^b) \quad (\text{V})$$

$$-\log \beta = \sigma(\log \bar{c} - \log \bar{c}) + \log(1 + \bar{r}^b) \implies 1 + \bar{r}^b = \beta^{-1} = 1 \implies \bar{r}^b = 0 \quad (\text{VI})$$

Subtracting (VI) from (V):

$$0 = E_t[\sigma \hat{c}_t - \sigma \hat{c}_{t+1} + \log(1 + r_{t+1}^b) - \log(1 + \bar{r}^b)]$$

$$\log(1 + r_{t+1}^b) - \log(1 + \bar{r}^b) \approx \frac{1}{1 + \bar{r}^b}(r_{t+1}^b - \bar{r}^b) = r_{t+1}^b$$

Because $\bar{r}^b = 0$, the percentage-deviation transformation $\hat{r}_t^b = (r_t^b - \bar{r}^b)/\bar{r}^b$ is undefined, so r_{t+1}^b enters in *levels*, not as a hat variable. The linearised bond Euler is therefore:

$$\boxed{\sigma \hat{c}_t = \sigma E_t \hat{c}_{t+1} + E_t r_{t+1}^b}$$

(4) Production function.

$$\log y_t = \log z_t + \alpha \log k_t + (1 - \alpha) \log \ell_t \quad (\text{VII})$$

$$\log \bar{y} = \log \bar{z} + \alpha \log \bar{k} + (1 - \alpha) \log \bar{\ell} \quad (\text{VIII})$$

(VII) – (VIII):

$$\boxed{\hat{y}_t = \hat{z}_t + \alpha \hat{k}_t + (1 - \alpha) \hat{\ell}_t}$$

(5) Wage.

$$\log w_t = \log(1 - \alpha) + \log y_t - \log \ell_t \quad (\text{IX})$$

$$\log \bar{w} = \log(1 - \alpha) + \log \bar{y} - \log \bar{\ell} \quad (\text{X})$$

(IX) – (X):

$$\boxed{\hat{w}_t = \hat{y}_t - \hat{\ell}_t}$$

(6) Rental rate.

$$\log r_t^k = \log \alpha + \log y_t - \log k_t \quad (\text{XI})$$

$$\log \bar{r}^k = \log \alpha + \log \bar{y} - \log \bar{k} \quad (\text{XII})$$

(XI) – (XII):

$$\boxed{\hat{r}_t^k = \hat{y}_t - \hat{k}_t}$$

(7) Government budget constraint.

We subtract the steady state from the original equation:

$$b_{t+1} - \bar{b} = (1 + r_t^b)b_t - (1 + \bar{r}^b)\bar{b} + g_t - \bar{g} - T_t + \bar{T}$$

Expanding $(1 + r_t^b)b_t$ around steady state (using $\bar{r}^b = 0$):

$$(1 + r_t^b)b_t \approx (1 + \bar{r}^b)\bar{b} + \bar{b}r_t^b + \bar{b}\hat{b}_t$$

where r_t^b enters in levels (since $\bar{r}^b = 0$ makes the hat transformation undefined). Substituting:

$$\bar{b}\hat{b}_{t+1} = \bar{b}\hat{b}_t + \bar{b}r_t^b + \bar{g}\hat{g}_t - \bar{T}\hat{T}_t$$

$$\boxed{\hat{b}_{t+1} = \hat{b}_t + r_t^b + \frac{\bar{g}}{\bar{b}}\hat{g}_t - \frac{\bar{T}}{\bar{b}}\hat{T}_t}$$

(8) Fiscal rule.

Taking logs of $T_t = \bar{T}\left(\frac{b_t}{\bar{b}}\right)^{\phi_{T,b}}$:

$$\log T_t = \log \bar{T} + \phi_{T,b} \log \left(\frac{b_t}{\bar{b}}\right)$$

$$\boxed{\hat{T}_t = \phi_{T,b} \hat{b}_t}$$

(9) Resource constraint.

We subtract the steady state from the original equation:

$$y_t - \bar{y} = c_t - \bar{c} + k_{t+1} - \bar{k} - (1 - \delta)k_t + (1 - \delta)\bar{k} + g_t - \bar{g}$$

$$\bar{y}\hat{y}_t = \bar{c}\hat{c}_t + \bar{k}\hat{k}_{t+1} - (1 - \delta)\bar{k}\hat{k}_t + \bar{g}\hat{g}_t$$

$$\boxed{\hat{y}_t = \frac{\bar{c}}{\bar{y}}\hat{c}_t + \frac{\bar{k}}{\bar{y}}\hat{k}_{t+1} - (1 - \delta)\frac{\bar{k}}{\bar{y}}\hat{k}_t + \frac{\bar{g}}{\bar{y}}\hat{g}_t}$$

(10) Exogenous shock: Technology.

$$\log z_t = (1 - \rho_z) \log \bar{z} + \rho_z \log z_{t-1} + \varepsilon_{z,t} \quad (\text{IX})$$

$$\log \bar{z} = (1 - \rho_z) \log \bar{z} + \rho_z \log \bar{z} \quad (\text{X})$$

(IX) - (X):

$$\boxed{\hat{z}_t = \rho_z \hat{z}_{t-1} + \varepsilon_{z,t}}$$

(11) Exogenous shock: Government spending.

$$\log g_t = (1 - \rho_g) \log \bar{g} + \rho_g \log g_{t-1} + \varepsilon_{g,t} \quad (\text{XI})$$

$$\log \bar{g} = (1 - \rho_g) \log \bar{g} + \rho_g \log \bar{g} \quad (\text{XII})$$

(XI) - (XII):

$$\boxed{\hat{g}_t = \rho_g \hat{g}_{t-1} + \varepsilon_{g,t}}$$

Q7: Dynare file and IRFs after a government spending shock

The attached file `assignment1main.m` implements the model, computes the steady state from the calibrated values above, and asks Dynare for first-order IRFs.

Main economic mechanisms after a positive government spending shock.

1. **Output rises on impact.** The negative wealth effect induced by higher government spending causes households to increase labour supply (see below), which raises output via the production function. In an RBC model, output is supply-determined.
2. **Labour rises.** The spending shock generates a negative wealth effect: households feel poorer (anticipating future tax liabilities) and respond by working more and consuming less.
3. **Consumption falls.** Government absorption uses real resources, and households anticipate the present value of future lump-sum taxes required to service the extra debt. This shows that the households have a Ricardian behavior.
4. **Capital falls gradually in this calibration.** Although higher labour temporarily raises the marginal product of capital, the resource cost of public spending dominates, so investment is crowded out and the capital stock declines over time.
5. **The bond rate increases.** Lower current consumption relative to expected future consumption implies a positive risk-free return via the Euler equation.
6. **Debt rises and taxes adjust gradually.** Because taxes respond only partially each period, the government initially issues debt; then taxes rise endogenously according to the fiscal rule and debt slowly goes back to the steady-state.
7. **Government spending rises exogenously and persistently.** Because $\rho_g = 0.95$, the shock is quite long-lived.

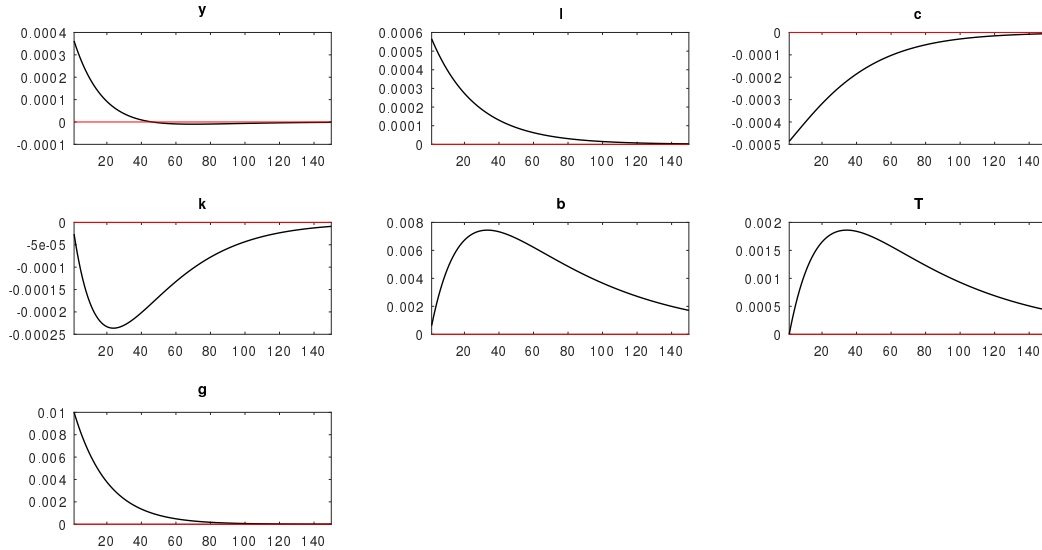


Figure 1: Shock in Public Spending (g_t)

Figure 1 shows the resulting IRFs from solving the linearised model for a one-standard-deviation government spending shock.

Q8. What happens if $\phi_{T,b}$ increases?

The mechanical effect on fiscal variables is immediate from

$$\hat{T}_t = \phi_{T,b} \hat{b}_t, \quad \hat{b}_{t+1} = \hat{b}_t + r_t^b + \frac{\bar{g}}{\bar{b}}(\hat{g}_t - \hat{T}_t).$$

A higher $\phi_{T,b}$ means:

- taxes react more aggressively to debt,
- debt accumulates less on impact,
- debt returns faster to target,
- the tax IRF is higher initially and less persistent later.

The non-trivial point is the effect on *real allocations*. In this model, Ricardian equivalence holds because: (i) taxes are lump-sum (non-distortionary), (ii) the representative household is infinitely-lived, (iii) there are no borrowing constraints, and (iv) the household and government discount at the same rate ($\beta(1 + \bar{r}^b) = 1$). As a result, debt is not net wealth, and changing the timing of taxes through $\phi_{T,b}$ leaves the present-value government budget constraint unchanged and therefore leaves the equilibrium paths of c_t , l_t , y_t , k_t , and (up to first order) the return path essentially unchanged.

So the correct answer is:

higher $\phi_{T,b}$ mainly changes the (b_t, T_t) dynamics, not the real allocation.

Q9. Would that answer change with a distortionary labour tax?

Yes, the conclusion changes.

If the government uses a labour-income tax τ_t instead of lump-sum taxes, the household budget contains net labour income $(1 - \tau_t)w_t l_t$, government revenues are $\tau_t w_t l_t$, and the tax rule is

$$\tau_t = \bar{\tau} \left(\frac{b_t}{\bar{b}} \right)^{\phi_{\tau,b}}$$

The intratemporal condition becomes

$$\chi l_t^\phi = c_t^{-\sigma} (1 - \tau_t) w_t$$

Now the fiscal rule directly changes the after-tax wage, hence labour supply, output, and investment. Therefore:

- a higher $\phi_{\tau,b}$ means labour taxes rise faster when debt increases,
- the labour wedge becomes larger,
- labour supply rises by less (and can even fall relative to the lump-sum benchmark),
- output and investment are lower,
- private activity is more strongly crowded out.

In short, with distortionary taxation the tax creates a *wedge* between the marginal rate of substitution and the marginal product of labour, making the timing of taxation matter for allocations. This breaks Ricardian equivalence: the speed of fiscal adjustment is no longer neutral.